



**COMSATS University Islamabad,
Sahiwal Campus.**

Daily Life Services Mobile Application

A Project Presented To

**COMSATS University, Islamabad
Sahiwal Campus**

In partial fulfillment

of the requirement for the degree of

Bachelor of Science in Software Engineering (2020-2024)

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The candidate confirms that the work submitted is their own and appropriate credit has been given where reference has been made to the work of other

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CERTIFICATE OF APPROVAL

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Executive Summary

The Sahulat App, an Android-based mobile application, emerges as a transformative solution in the daily life services. In an era defined by convenience and instant access, Sahulat App revolutionizes the traditional approach to service acquisition. Designed to seamlessly connect users with skilled workers, this app transcends the conventional methods of appointment booking for essential services, such as plumbing, electrical work, cleaning, and more.

The primary goal of the Sahulat App is to provide a user-friendly platform where individuals and organizations can effortlessly access a diverse array of services without the need for in-person meetings or time-consuming phone calls. By prioritizing convenience, security, and efficiency, Sahulat App bridges the gap between service seekers and service providers.

This innovative platform not only simplifies the lives of users but also extends a unique opportunity for workers to showcase their expertise and earn income through the app. The Sahulat App leverages cutting-edge technologies, including Flutter, Dart, and Firebase, to deliver a comprehensive ecosystem where users and workers can build their portfolios.

The analysis from our literature review highlights that many existing systems are limited to serving individual homeowners. In contrast, Sahulat App extends its services to cover homes, offices, hospitals, and schools, demonstrating its versatility and broad reach. Additionally, the app introduces a novel feature that allows workers to sign up and earn money, setting it apart from existing systems.

Employing the incremental model for development, Sahulat App ensures that each phase of the project is meticulously designed, developed, tested, and refined. This iterative approach enables ongoing enhancements, bug fixes, and updates in response to user feedback.

Sahulat App will emerge as a game-changer in the realm of daily life services, offering a secure, efficient, and user-centric solution for connecting service seekers with providers. With a focus on convenience, transparency, and versatility, Sahulat App will revolutionize how we access essential services in today's fast-paced world.

Acknowledgement

All praise is to Almighty Allah who bestowed upon me a minute portion of His boundless knowledge by virtue of which I was able to accomplish this challenging task.

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Abbreviations

SDLC	Software Development Life Cycle
API	Application Programming Interface
SDK	Software Development Kit
SRS	Software Require Specification
SDD	Software Design Description
GUI	Graphical User Interface
FR	Functional Requirements
NFR	Non-Functional Requirements
DFD	Data Flow Diagram

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Chapter 1

Introduction

1. Introduction

In this time when everyone wants to save their time and wants their work to be done as fast as possible. It is very difficult for everyone to go out and find worker for their desired service. Whether you're seeking assistance for a specific task or looking to provide your services, Sahulat App streamlines the process. It acts as a seamless bridge connecting users with workers, eliminating the need for time-consuming searches and inquiries. This innovative application prioritizes user convenience and expediency, addressing the contemporary need for efficient service access.

1.1 Brief

We are trying to make an App that serves as a seamless platform connecting users with workers, cut across the traditional methods of appointment booking for plumbing, electrical, cleaning, and other essential services. With the click of a button, users gain access to a vast network of skilled professionals, eliminating the need for in-person meetings or phone calls. Moreover, Sahulat App goes beyond mere convenience – it prioritizes security, ensuring that users can request services without having to visit the physical location of the worker. This added layer of safety is particularly crucial in today's fast-paced and ever-changing world. We believe that everyone deserves to have essential services delivered to their doorstep with ease, and the Sahulat App makes that a reality. Sahulat App not only simplifies the lives of users but also empowers skilled workers by providing them with a unique opportunity to showcase their expertise and earn income through the platform. We believe in creating a symbiotic relationship between users and workers, fostering trust and collaboration. It is powered by modern technology, including Flutter, Dart, and Firebase, Sahulat App is meant to be a single platform providing an ecosystem where users and workers can build their portfolios.

1.2 Relevance to Course Modules

The Sahulat App project aligns closely with a Bachelor's degree in Software Engineering by primarily focusing on mobile application development for the Android platform. It directly aligns with core software engineering principles, emphasizing skills related to mobile app design, user interface development, database management, and software implementation. This project serves as a practical embodiment of software engineering concepts, showcasing our proficiency in creating a user-centric Android application using modern technologies like Flutter and Dart. Our academic journey has provided us with a solid foundation in software engineering, making us well-equipped to undertake this project and apply our knowledge effectively.

1.3 Project Background

The project, Sahulat App, originates from the need to modernize and simplify the process of accessing daily life services. In today's rapidly advancing technological landscape, the conventional approach of scheduling appointments with workers, such as plumbers, electricians, and cleaners, can be inefficient. Sahulat App aims to revolutionize this system by offering a mobile Android application that connects users with workers. The app eliminates the need for physical visits or time-consuming phone calls, enhancing user convenience.

Through the Sahulat App, users can effortlessly request various services at their convenience, and the app matches them with skilled professionals in their vicinity. This innovative approach not only streamlines daily service requirements but also provides a unique opportunity for workers to showcase their expertise and earn income through the platform. Sahulat App use cutting-edge technologies, including Flutter, Dart, and Firebase, to deliver an intuitive user interface, instant service accessibility, and a dependable solution for connecting service seekers with providers.

1.4 Literature Review

There are many existing apps that only focus on helping individual homeowners. In contrast, the Sahulat App project goes further by addressing this limitation. It doesn't just serve homes but also offers services for offices, hospitals, and schools. This expansion aligns with the project's goal of providing a wide range of secure services to different places, making it more versatile.

Table 1.1 Related System Analysis

Application Name	Weakness	Proposed Project Solution
Karsaaz App (Professional Services Mobile Application)	For this system, they do not provide a chat feature that enables users to negotiate prices directly with professional workers.	Our application provide a real-time chat option, allowing users to communicate directly with professional workers.

1.5 Analysis from Literature Review

The analysis of the literature review shows that many existing systems only focus on helping individual homeowners. In contrast, the Sahulat App project goes further by addressing this limitation. It doesn't just serve homes but also offers services for offices, hospitals, and schools.

This expansion aligns with the project's goal of providing a wide range of secure services to different places, making it more versatile.

The Sahulat App introduces a feature by extending its services to workers. This unique opportunity for skilled professionals to sign up and offer their services not only enriches the platform with a diverse range of expertise but also creates an inclusive ecosystem.

1.6 Methodology and Software Lifecycle for This Project

We have selected incremental model for this project. Waterfall model has many drawbacks in the development of any system, to overcome the drawbacks incremental model is used in the development phase of our project. In a model the system is developed in increments and in module wise one after one, each modules contains more functionalities than before. These smaller pieces then built and delivered to management in increments. This iterative approach minimizes the risk of late-stage surprises and increases the likelihood of delivering a product that truly meets the expectations of all parties involved. The incremental model is the perfect fit for our project's development phase, offering flexibility, adaptability, and responsiveness. It's a methodology that ensures we can navigate the complexities of software development with confidence and deliver a product that not only meets but exceeds expectations.

1.6.1 Rationale behind Selected Methodology

We are using the incremental model because it divides the software into submodules and each sub module is developed by following the software development life cycle process SDLC like Analysis, Design, Code, and Test. Easier to test and debug during a smaller iteration. We can implement our project phase wise that's why it is very helpful using the incremental model in our project.

As the proposed system is based on a new idea and there is a lot of unclear and complex information. The reasons for using this methodology are:

1. Because it is a new software so new features will be added in the future, and this methodology supports this feature very well.
2. It is acceptable to tackle difficult problems first. It allows effective management of software complexity under.
3. The model fosters a collaborative and adaptive development environment. As new features are seamlessly integrated, the methodology supports our commitment to delivering a user-centric app.
4. By tackling difficult problems early on, the incremental model contributes to effective software complexity management.

Chapter 2

Problem Definition

2. Problem Definition

2.1 Problem Statement

To take the appointment of worker we have to call him or with the personal meeting we can meet him, and it is not sure that we get the appointment of the worker. It is also not secure in terms of safety concern. People feel difficulty in taking the appointment of worker. They have to call or meet them personally to have an appointment. It is not sure that they get the appointment and there are many problems occur, like the worker is busy at somewhere else or he is not present at his office etc.

The same system exist already in different countries and in bigger cities within country, but it is not currently working in our city. The other system only covers household services but we take this system to next level by giving same services to hospital, organizations, offices etc.

After completing this project, we have getting Hands on Experience of Full Stack Android Developer with Flutter Framework and Dart with Firebase Database.

2.2 Deliverables and Development Requirements

The project's deliverables encompass the following key components:

- i. **Sahulat Mobile App:** The primary deliverable is the fully functional Sahulat App, designed for Android devices. This app will enable users to access a wide range of daily life services and connect user with workers.
- ii. **User Registration Interface:** A user registration system will be implemented to allow users to create accounts, manage their profiles, and access the app's features securely.
- iii. **Worker Registration Interface:** A dedicated portal for workers to sign up, provide their service details, and become part of the Sahulat App ecosystem.
- iv. **Database Management:** The project involves setting up and managing a secure database system, which will store user profiles, worker profile, worker details and service listings.
- v. **User Interface Design:** A user-friendly and intuitive interface will be developed to enhance the overall user experience. Our team is dedicated to crafting an interface that not only prioritizes user-friendliness but also delights users with its intuitive design. Our goal is to elevate the overall user experience to new heights.
- vi. **Admin Dashboard:** An admin dashboard will be created to oversee app activities, manage users and workers, and ensure the smooth operation of the platform.
- vii. **Testing and Quality Assurance:** Rigorous testing procedures will be conducted to identify and rectify any bugs or issues, ensuring a stable and reliable app.
- viii. **Development Requirements:** In the foundation for the Sahulat App, the choice of

development tools is pivotal to ensure seamless functionality and a user-friendly interface. The development environment is anchored by Android Studio and SDK, providing a robust framework for Android app development.

- **Android Development Environment:** Android Studio and SDK for app development.
- **Programming Languages:** Utilization of Flutter for cross-platform app development and Dart for coding.
- **Backend Services:** Firebase for database management and hosting.
- **User Interface Design Tools:** Figma tool for designing the app interface.

2.3 Current System

The current systems for similar service systems may vary from country to country and even within major cities of our own country. However, it's crucial to highlight that while these systems might thrive in some regions, they remain notably absent in our specific city. While the existing systems predominantly target household services, our Sahulat App is poised to take this concept to a new pinnacle of service excellence by expanding its reach to encompass not just homes but also hospitals, organizations, and offices.

Our application provides an extraordinary opportunity for skilled workers to register and offer their valuable services, culminating in the creation of an all-encompassing ecosystem that serves both those seeking services and those providing them.

The Sahulat App stands apart from existing systems by breaking free from the constraints of a purely household-focused model. Its adaptability extends across a wide spectrum of sectors, recognizing the multifaceted service requirements of individuals and organizations alike. Our app's versatility knows no bounds, offering a comprehensive solution tailored to the diverse needs of residents and businesses in our city.

In essence, the Sahulat App is not just a service platform; it's a catalyst for change, bridging the existing service gap in our city and tailoring solutions to address the specific demands of our local community. We're committed to making a positive impact, one service at a time.

CHAPTER 3

Requirement Analysis

3. Requirement Analysis

3.1 Use Case Diagram

In this Use case diagrams, we show the behavior of our system and frequently use it to analyze various systems. They enable you to visualize the different types of roles in a system and how those roles interact with the system.

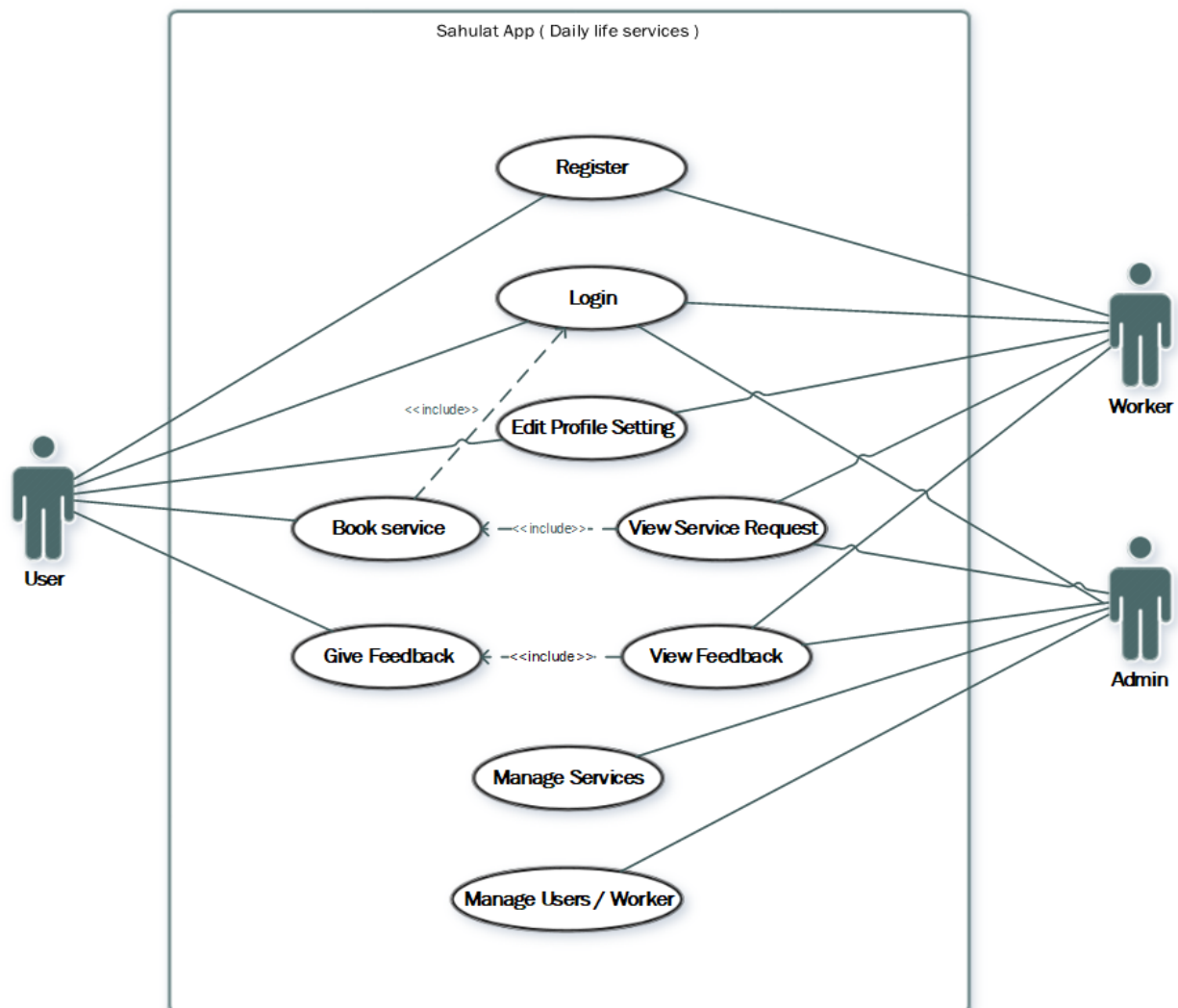


Figure 3.1 Use Case Diagram

The Sahulat App's use case diagram illustrates key interactions between actors and the system. Users can register, request services, and provide feedback, while workers interact by accepting or declining service requests. Administrators manage bookings and worker information. This comprehensive use case diagram encapsulates the essential functionalities, promoting efficient collaboration among users, workers, and administrators within the Sahulat App ecosystem.

3.2 Use Case Descriptions

3.2.1. Register Use Case

Table 3.1 Register Use Case

Use Case ID:	UC-111
Use Case Name:	Register
Actors:	Primary Actor: User, Worker, Secondary Actor: Admin
Description:	A user accesses the Sahulat App, and register its account on App. A worker register the Sahulat App by giving service details. An admin is responsible for all verification and overlooks of the registration by users and workers.
Trigger:	The user indicates that he wants to register to an App. A worker indicates that he is wants to register to give services. An Admin is responsible for all maintenance and overlooks of work.
Preconditions:	User PRE-1. Users must have an internet connection PRE-2. User must need to fill all essential fields. Worker PRE-1. Worker must have an internet connection PRE-2. Worker must need to fill all essential fields.
Post conditions:	User POST-1. User data must be created and saved in the database. All necessary fields must be filled. The system displays the login page. Worker POST-1. Worker data must be created and saved in the database. All necessary fields must be filled. The system displays the login in page.
Normal Flow:	User 1.The user indicates a desire to create account and enter basic information. 2.System creates a new user. 3.The user fills one or more input fields. 4.All fields must be saved. Worker 1.The worker indicates a desire to create account and enter basic information. 2.System creates a new user. 3.The worker fills one or more input fields. 4.All fields must be saved.

Alternative Flows:	There is not alternative flow for account registration.
Exceptions:	User 1. Basic user data is incomplete. 2. User account already exists Worker 1. Basic user data is incomplete. 2. Worker account already exists

3.2.2. Login Use Case

Table 3.2 Login Use Case

Use Case ID:	UC-112
Use Case Name	Login
Actors:	Primary Actor: User, Worker, Secondary Actor: Admin
Description:	A user enters fields to log in. A worker enters fields to log in. An admin is responsible for all verification and oversight of the login by users and workers.
Trigger:	The user indicates that he wants to login to an App. A worker indicates that he is wants to login to give services. An Admin is responsible for all maintenance and oversight of work.
Preconditions:	User PRE-1. Users must have an internet connection. PRE-2. User must have an account to login. PRE-3. User must need to fill all essential fields. Worker PRE-1. Worker must have an internet connection. PRE-2. Worker must have an account to login. PRE-3. Worker must need to fill all essential fields.
Post conditions:	User POST-1. User login data must be verified from the database. All necessary fields must be filled. The system displays the login page. Worker POST-1. Worker data must be created and saved in the database. All necessary fields must be filled. The system displays the login page.

Normal Flow:	User 1. The user indicates a desire to login to account. 2. System verifies the user. 3. The user fills email and password input fields. 4. All fields must be saved. Worker 1. The worker indicates a desire to login to account. 2. System verifies the worker. 3. The worker fills email and password input fields. 4. All fields must be saved.
Alternative Flows:	Password incorrect, he/she needs to change password by generating a new password by email verification.
Exceptions:	User 1. User account does not exist. 2. User email/password is incorrect, he/she needs to forget password. Worker 1. Worker account does not exist. 2. Worker email/password is incorrect, he/she needs to forget password.

3.2.3. Edit Profile Setting Use Case

Table 3.3 Edit Profile Setting Use Case

Use Case ID:	UC-113
Use Case Name	Edit Profile Setting
Actors:	Primary Actor: User, Worker
Description:	A user and worker can edit username, change email, change password, change profile picture.
Trigger:	The user indicates that he wants to edit profile setting. The worker indicates that he wants to edit profile setting.
Preconditions:	User PRE-1. User must be log into the account. Worker PRE-1. Worker must be log into the account
Post conditions:	User POST-1. User can edit username, change profile picture, email and password. Update Successfully pop up will be shown. Worker

	POST-1. Worker can edit username, change profile picture, email and password. Update Successfully pop up will be shown.
Normal Flow:	User 1. User need to login in an account. 2. User can edit username. 3. User can change profile picture. 4. User can change email. 5. User can change password. Worker 1. Worker need to login in an account. 2. Worker can edit username. 3. Worker can change profile picture. 4. Worker can change email. 5. Worker can change password.
Alternative Flows:	User can change any profile setting or more than many one setting change. Worker can change any profile setting or more than many one setting change
Exceptions:	User 1. User account not exists. 2. User not login to the app. Worker 1. Worker account not exists. 2. Worker not login to the app.

3.2.4. Book Service Use Case

Table 3.4 Book Service Use Case

Use Case ID:	UC-114
Use Case Name	Book Service
Actors:	Primary Actor: User
Description:	A user can book any service or book more than one service showing in the app. User will select any service, enter details (date, time) and book service.
Trigger:	The user indicates that he wants to book service.

Preconditions:	PRE-1. User must be log into the account.
Post conditions:	POST-1. Pop-up will be shown that your booking is successfully done. POST-2. He/she will get SMS in his/her sim. POST-3. He/she will get email from an app for booking.
Normal Flow:	1. User need to login in an account. 2. User can select any service showing in homepage. 3. User have to select address. 4. User have to select date/ time. 5. User have to click on place order.
Alternative Flows:	User can go to services page to view more services and book any of them.
Exceptions:	1. User account not exists. 2. User not login to the app. 3. User didn't have any address details.

3.2.5. View Service Request

Table 3.5 View Service Request

Use Case ID:	UC-115
Use Case Name	View service request
Actors:	Primary Actor: Worker
Description:	A worker can view booked services and manage all bookings such as worker can cancel request and confirm request.
Trigger:	The worker indicates that he wants to view and manage requested services.
Preconditions:	PRE-1. Worker must be logged into the account. PRE-2. There should be requested services on the booking page.
Post conditions:	POST-1. A pop-up will be shown that request has been confirmed or canceled. POST-2. The user will receive a confirmation message on the app if their request is accepted or not. POST-3: Request details will be displayed on the canceled screen if the request is canceled or confirmation screen if accepted.
Normal Flow:	1. Worker needs to log in to an account. 2. Worker can select any requested service. 3. Worker can delete request. 4. Worker can confirm request.

Alternative Flows:	Worker can also view services request on a specific service page.
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3.2.6. Give Feedback

Table 3.6 Give Feedback

Use Case ID:	UC-116
Use Case Name	Feedback
Actors:	Primary Actor: User, Admin
Description:	A user can give feedback to admin about app. An admin can view feedback.
Trigger:	The user indicates that he wants to give feedback of app to an admin. An admin indicates that he is wants to view feedback from users.
Preconditions:	User PRE-1. Users must have an internet connection PRE-2. User must login to the app. Admin PRE-1. Admin must have an internet connection PRE-2. Admin must login to the app.
Post conditions:	User POST-1. User feedback will be send to admin in message section. Admin POST-1. Feedback must be displayed on admin message section.
Normal Flow:	User 1. The user indicates a desire to give feedback. 2. System store it to database. 3. All fields must be saved. Admin 1. The admin indicates a desire to view feedback given by different users. 2. System display users' feedback to admin screen. 3. All fields must be saved.

3.2.7. Manage Service

Table 3.7 Manage Service

Use Case ID:	UC-118
Use Case Name:	Manage Services

Actors:	Primary Actor: Admin
Description:	An admin is responsible for managing all services. Services includes: electrician, plumbing etc., Admin can add, delete and edit services and services details.
Trigger:	The admin indicates that he wants to manage all services of app. An Admin is responsible for managing all services.
Preconditions:	PRE-1. Admin must have an internet connection or linked with database. PRE-2. Services data must be existed in database.
Post conditions:	POST-1. Service and Services' details add, delete and modified in database.
Normal Flow:	1. Admin login to his page. 2. View services details on services page. 3. Admin select add, delete and modified button and confirm.

3.2.8. Manage User / Worker

Table 3.8 Manage User / Worker

Use Case ID:	UC-117
Use Case Name:	Manage Users/ Worker
Actors:	Primary Actor: Admin
Description:	An admin is responsible for managing all users and worker. It include app user: clients and workers. The admin can add, delete, and edit users and user details.
Trigger:	The admin indicates that they want to manage all users/ worker of the app. An admin is responsible for managing all users/ worker.
Preconditions:	PRE-1. Admin must have an internet connection or be linked with the database. PRE-2. The user/ worker must exist in the database.
Post conditions:	User/ worker and their details are added, deleted, or modified in the database.
Normal Flow:	1. Admin logs in to their page. 2. Views users' details on the user page. 3. Admin selects add, delete, or modify button and confirms.

3.3 Functional Requirements

The following modules will be included in this system:

- Module 1: User Registration Module
- Module 2: Profile Management Module
- Module 3: Service Request and Booking Module
- Module 4: Real-time Chat Module
- Module 5: Worker Verification and Authentication Module
- Module 6: Notification Management Module

All modules will be in android applications that can only be accessed via mobile devices.

3.3.1. User Registration Module

1. Users can create a new account by providing a valid email address and password.
2. The system should validate the uniqueness of the email address.
3. Users receive a verification email for account activation.
4. A "Forgot Password" functionality is available for password recovery.

3.3.2. Profile Management Module

1. Users can view and edit their personal information, including name, contact details, and address.
2. Users can upload, change, or remove their profile picture.
3. Workers can create and manage professional profiles with details such as skills, certifications, and work experience.

3.3.3. Service Request and Booking Module

1. Users can browse available services categorized by type.
2. Users can select a specific service and view details, including service description and cost.
3. Users initiate service requests by providing details such as preferred date, time, and location.
4. Users can review and confirm booking details before finalizing the request.
5. The system sends booking confirmation notifications to users and workers involved.

3.3.4. Real-time Chat Module

1. Users and workers can initiate real-time chat for discussing service details.

3.3.5. Worker Verification and Authentication Module

1. Workers must submit evidence of their expertise, including certifications, licenses, or portfolios.
2. The system verifies the authenticity of provided documents before approving the worker.
3. The system checks for any legal or professional issues related to the worker.

3.3.6. Notification Management Module

1. Users receive notifications for booking requests, confirmations, and cancellations.
2. Users receive general notifications for app updates, promotions, and announcements.

3.3.7. Admin Module

Admin access is allowed only to admin or any business man who will run its business using this app. Admin can make modification in the database and backend. Such as adding, deleting and updating services and workers from the database record.

3.4 Non-Functional Requirements

Non-functional Requirements are those that establish parameters that may be used to evaluate the performance of a system. Non-functional requirements are the limitations to which the system will work.

3.4.1 Usability

The system will assist the user in locating and obtaining various services in the shortest amount of time. The consumer can choose between multiple workers or selective services, which enhances service availability.

3.4.2 Accessibility

Anyone can access this app, but the main functionalities can only be accessed by mentioned city (Sahiwal).

3.4.3 Responsiveness

The system replies immediately when the user have any query regarding services or booking. The system responds to every user error. When a customer orders a service, this system immediately sends the client's information to the worker.

3.4.4 Availability

This mobile application will be available to all users at all times; however, service facilities will only be provided during business hours (8AM – 8PM).

3.4.5 Scalability

The Sahulat App is designed to scale seamlessly to accommodate a growing user base and an expanding range of services. The architecture is flexible, allowing for efficient scaling to meet increasing demands without compromising performance.

Chapter 4

Design and Architecture

4. Design and Architecture

4.1 System Architecture

The structure of the system explains the working of the app and describes the components of the project, their relationships, and how they interact with each other, with this architecture. It intricately outlines the working mechanism of the app, detailing the relationships between its various components. This comprehensive architecture elucidates how different modules and functionalities seamlessly integrate, ensuring a cohesive and efficient operation.

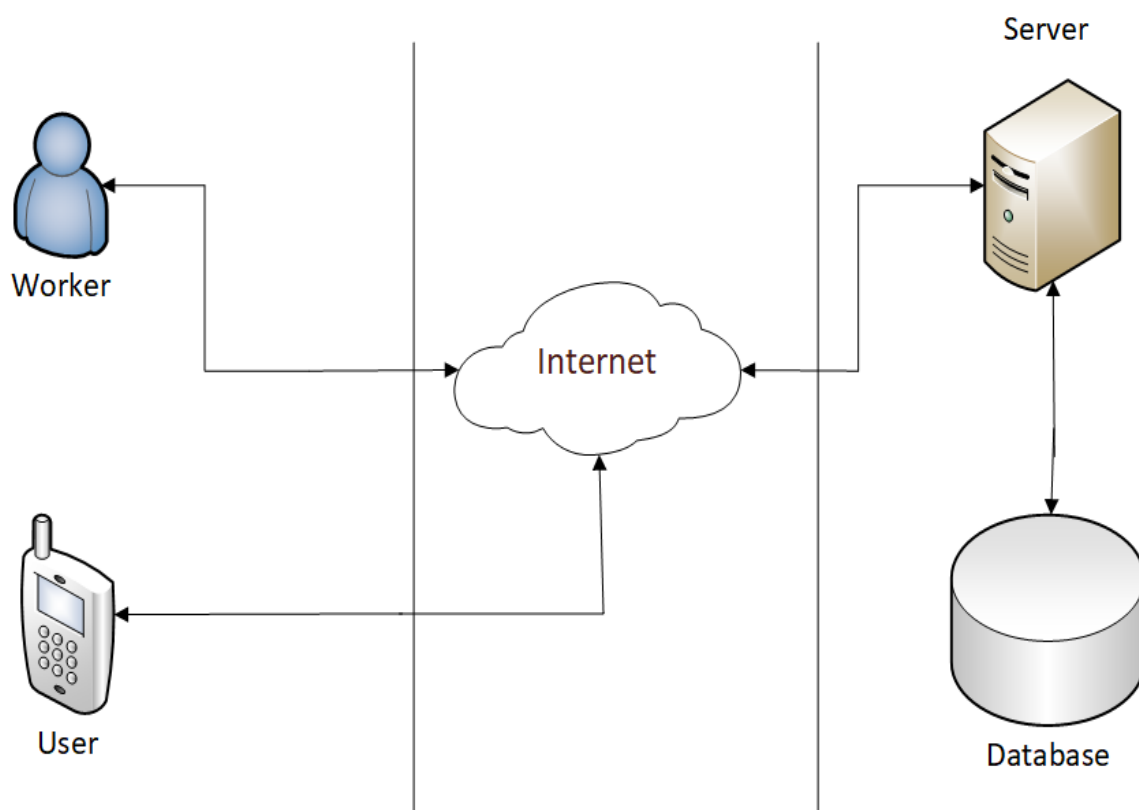


Figure 4.1 Architectural Diagram

This architecture not only provides a roadmap for developers but also serves as a foundation for scalability and adaptability. It emphasizes the modularity and flexibility of the system, allowing for future enhancements and the incorporation of new features. By visualizing the relationships and dependencies, the system architecture serves as a vital tool in ensuring a robust, reliable, and user-centric application that aligns with the project's overarching goals and requirements.

4.2 Process Flow/Representation

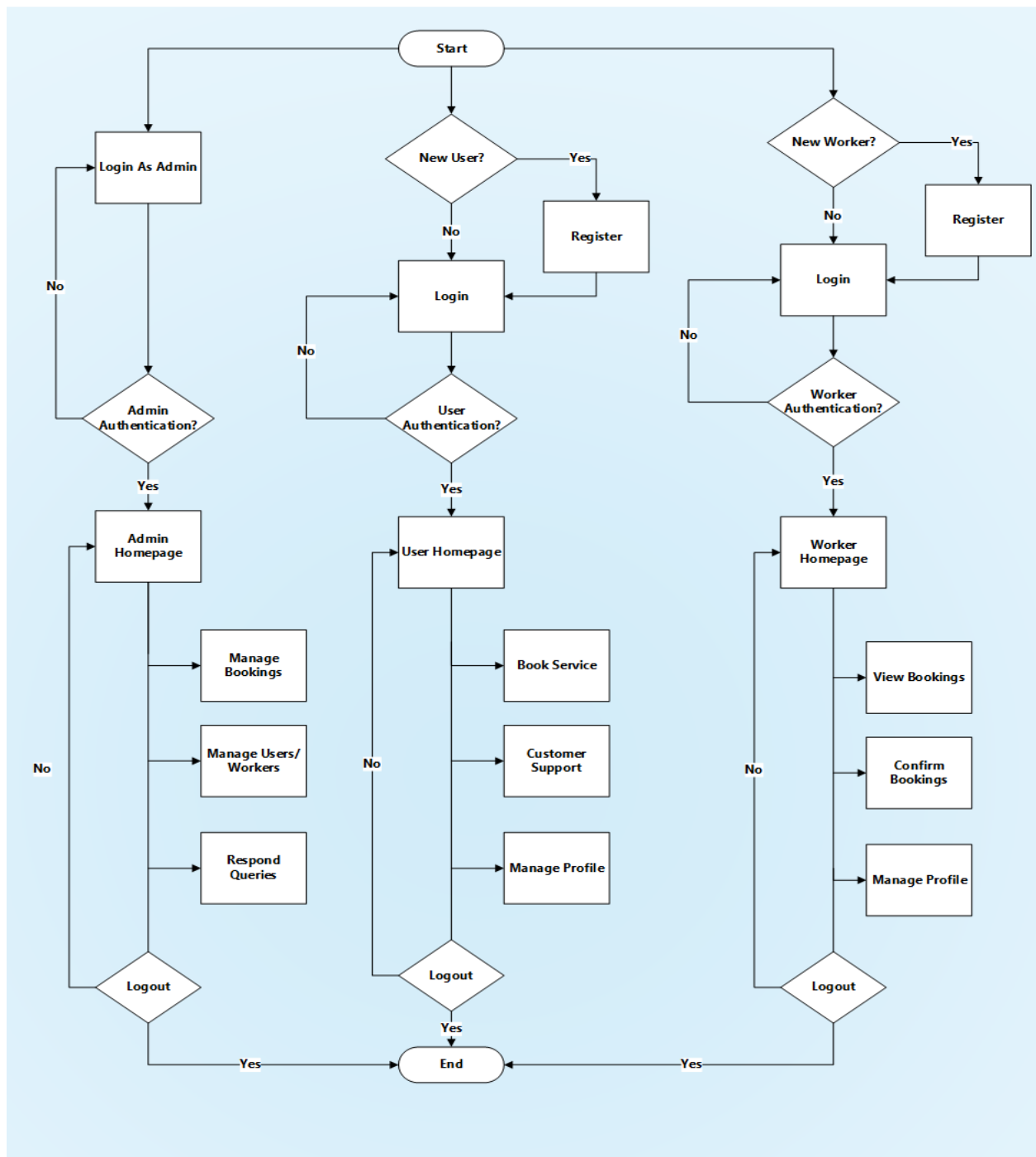


Figure 4.2 Process Flow

The process flow diagram for the Sahulat App delineates a systematic sequence of actions. It begins with user registration, leading to service requests and feedback submission. Simultaneously, workers receive and respond to service requests. Administrators oversee and manage bookings. The flow ensures a streamlined process, allowing users to access services seamlessly, workers to efficiently handle requests, and administrators to maintain operational oversight.

4.3 Class Diagram

Class diagram provides a visual representation of the relationships and interactions between different components of the Sahulat App, fostering a clear understanding of the system's structure and functionality.

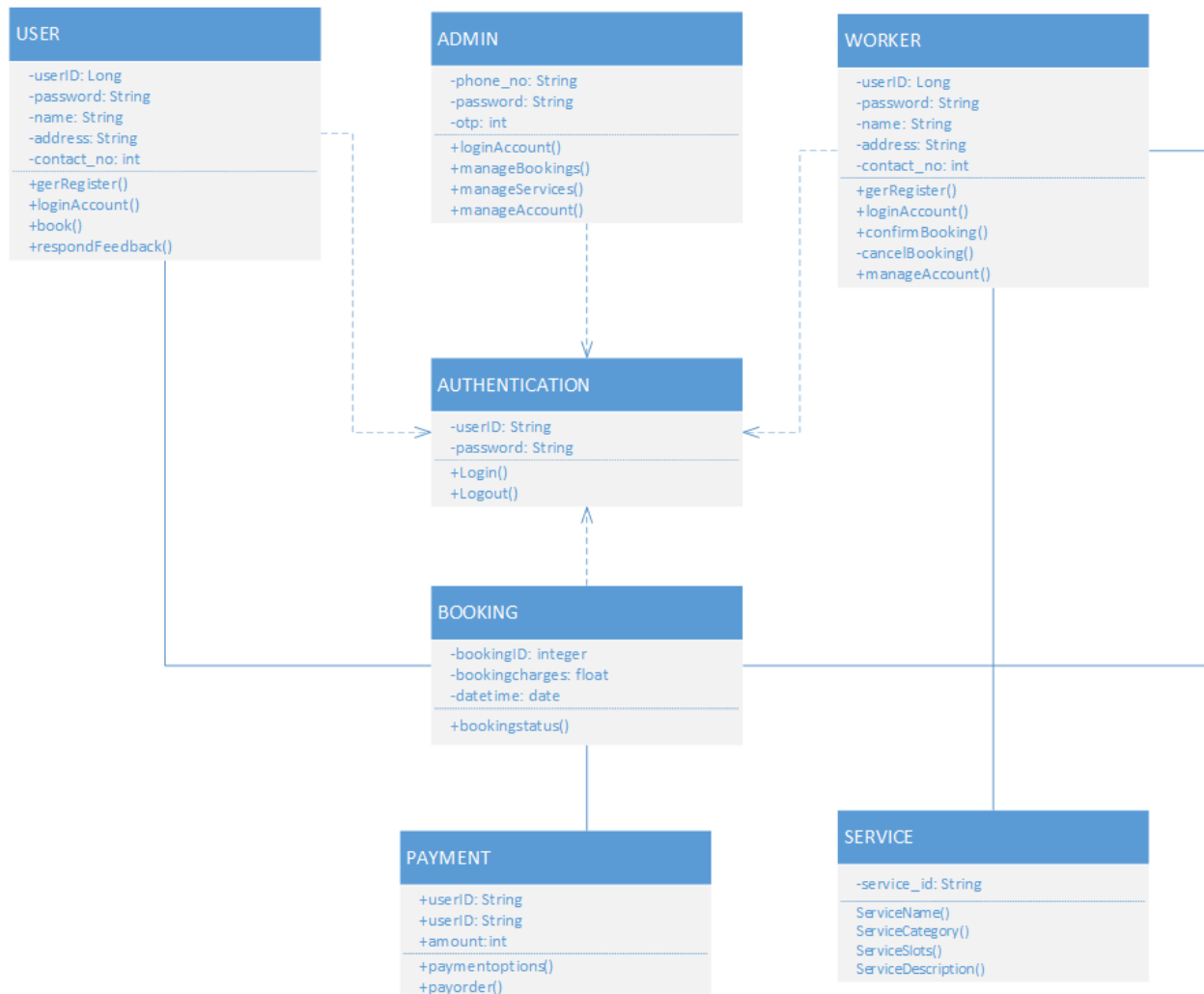


Figure 4.3 Class Diagram

In the class diagram, fundamental classes embody the core functionalities of the system. The User class encapsulates user-related attributes and interactions, while the Admin class oversees administrative functions. Skilled workers are represented by the Worker class, which interacts with the Booking class to manage service requests. The Authentication class ensures secure access, and the Payment class handles financial transactions. The Booking class orchestrates service requests, connecting with the User, Worker, and Payment classes, while the Service class defines the various services available.

4.4 Sequence Diagram

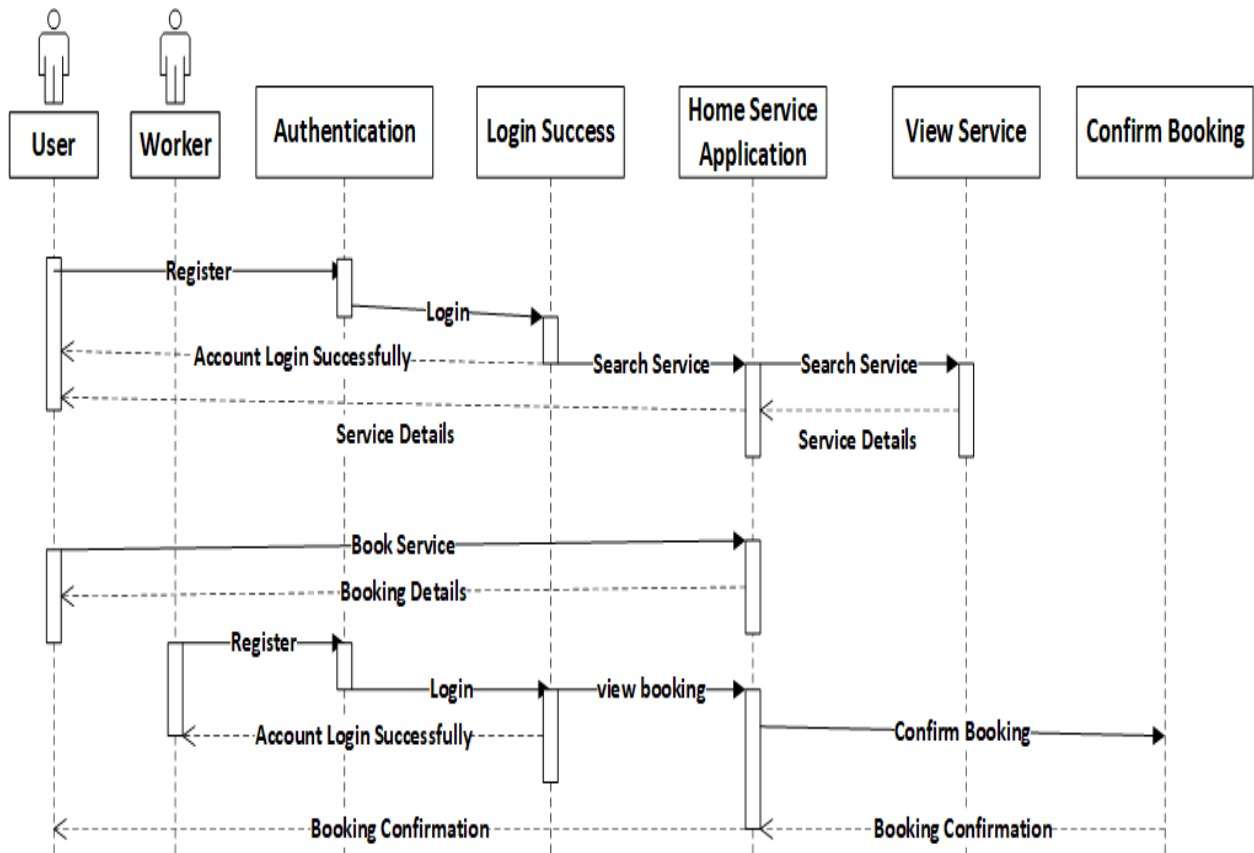


Figure 4.4 Sequence Diagram

The sequence diagram for the Sahulat App offers a detailed representation of interactions among various actors within the system.

- **User Interaction**

The sequence begins with a user registering, triggering a message to the system for authentication. Once authenticated, the user proceeds to submit a service request, initiating another message to the system. Simultaneously, the system validates the user's request and logs the service request.

- **Worker Interaction**

Workers authenticate their credentials, entering the "Accepting Requests" phase. As service requests come in, workers review and respond.

- **Booking Confirmation**

Upon a worker's acceptance of a service request, the system updates the booking status and notifies the user, leading to a confirmation message.

4.5 State Transition Diagram

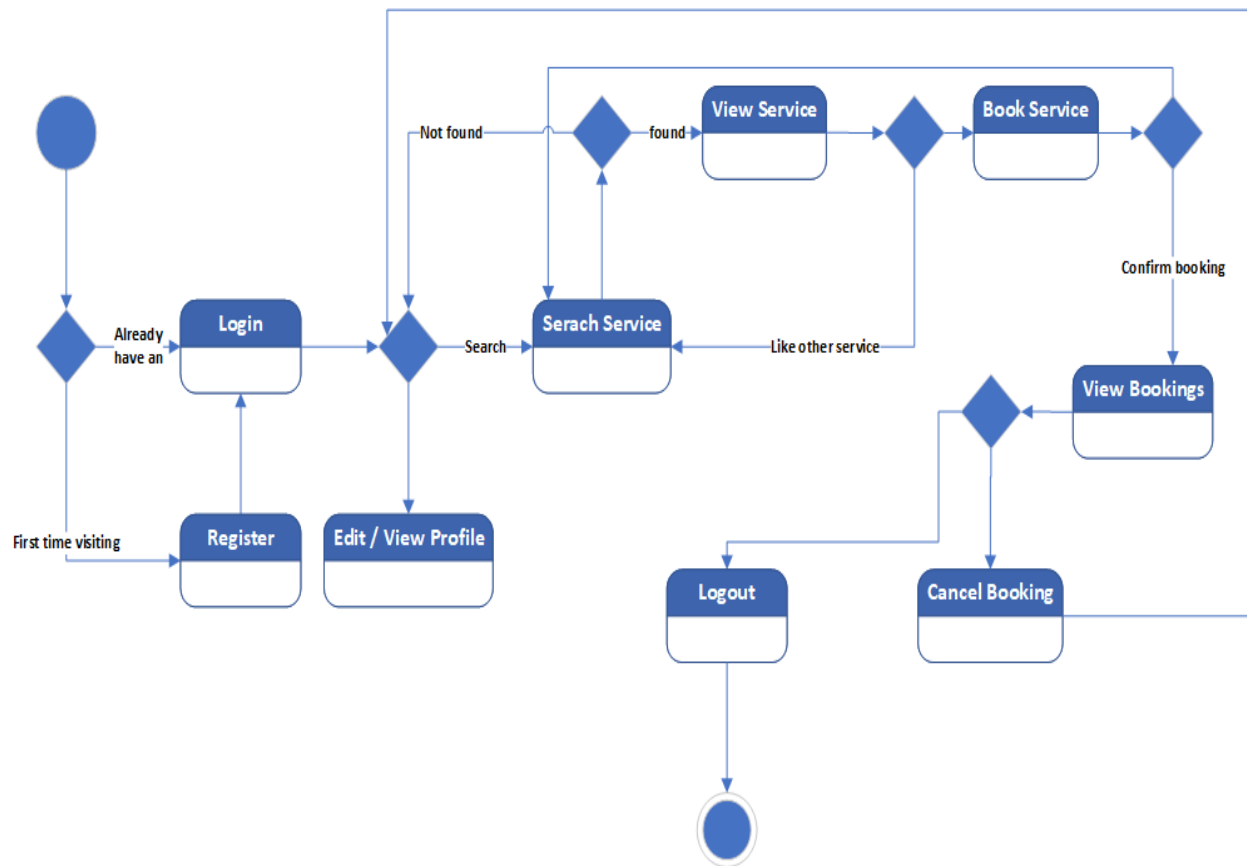


Figure 4.5 State Transition Diagram

The state transition diagram for the Sahulat App encapsulates key states and transitions.

- **User States**

Users initially enter the system in this state after completing the registration process. Upon initiating a service request, users transition to this state, indicating their active engagement in the service-seeking process. After a worker accepts the service request, the user transitions to this state.

- **Worker States**

Workers enter the system in this state after successful authentication. In this state, workers actively review and respond to service requests, deciding whether to accept or decline.

- **Admin States**

Administrators, responsible for overseeing bookings, enter this state to manage and coordinate service assignments. Administrators may transition to this state when updating worker information or making system-wide adjustments.

4.6 Data Flow Diagram

4.6.1 DFD Level 0

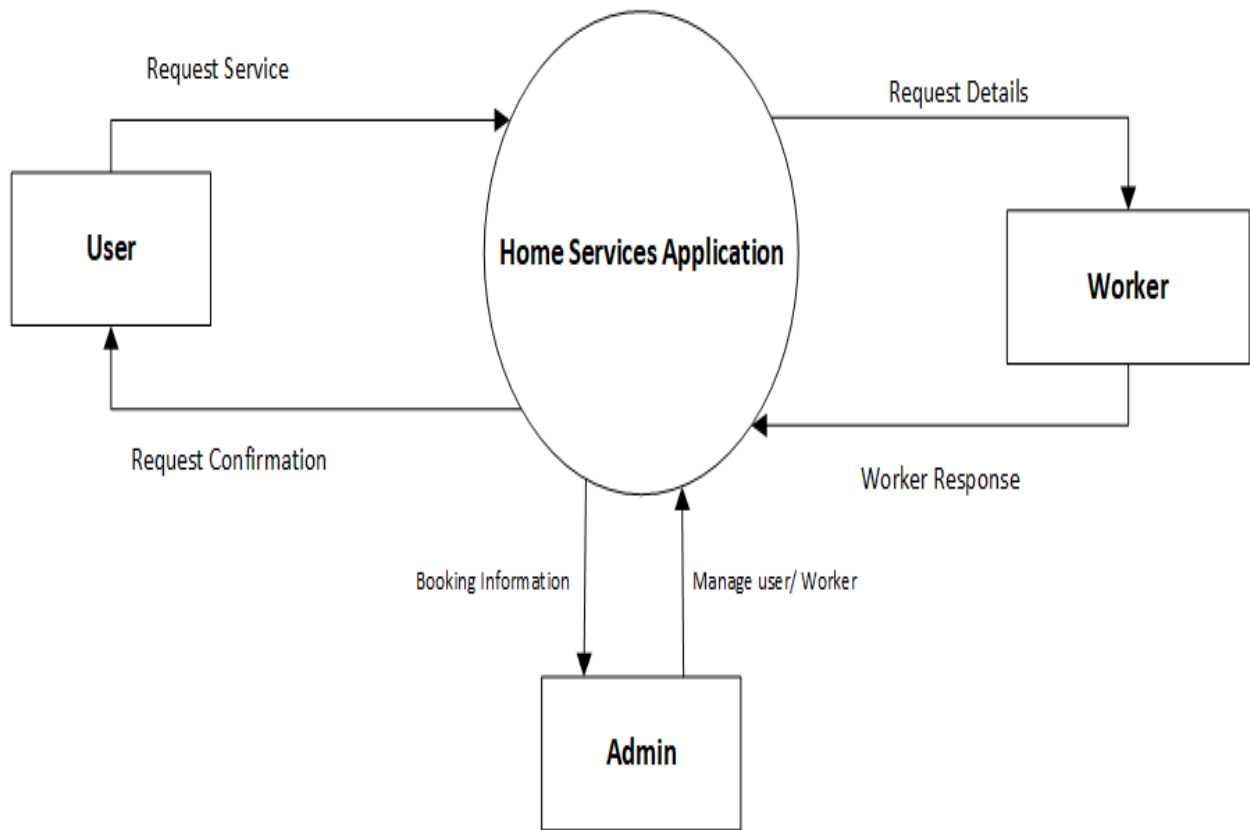


Figure 4.6 DFD Level 0

In the Data Flow Diagram, the primary entities and their interactions are succinctly illustrated. The process initiates with users, who request services by providing relevant details such as service type, preferred time, and location. This user-initiated data then flows to the Worker entity, representing skilled workers on the platform. Workers, in turn, receive and respond to service requests based on their expertise and availability. Simultaneously, the service request information is transmitted to the Admin entity, signifying the administrative role in overseeing and managing bookings. Admins play a crucial role in coordinating, confirming, and optimizing service assignments based on worker availability and user preferences. This high-level DFD Level 0 encapsulates the fundamental data flow and interactions between users, workers, and administrators, offering a comprehensive overview of the Sahulat App's essential functionalities.

4.6.2 User DFD Level 1

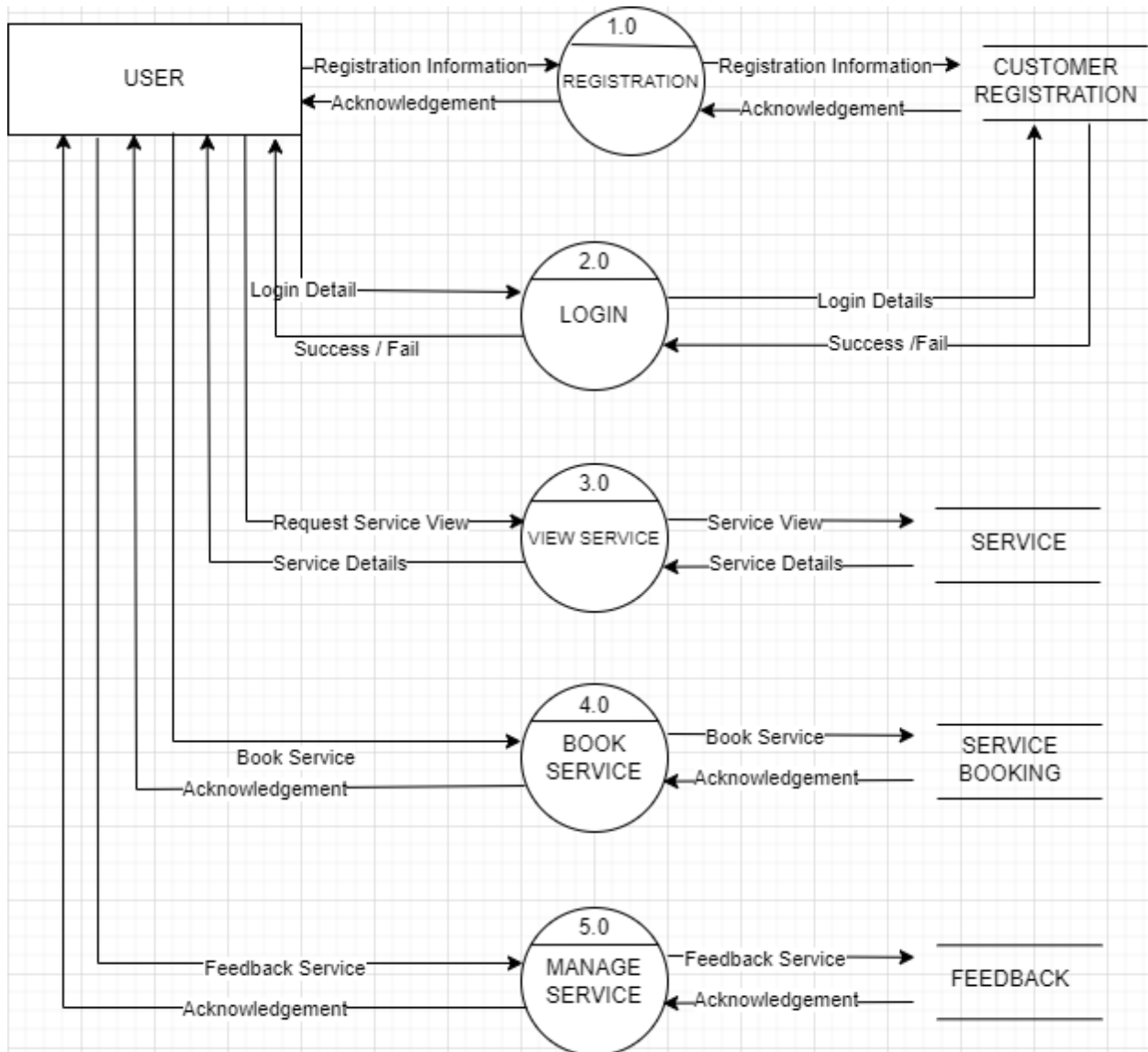


Figure 4.7 User DFD Level 1

At DFD Level 1 for the User entity, the detailed interactions and data flows are unveiled. Users, after initiating a service request, provide specific details such as service preferences, location, and desired time through the app interface. This user-input data is then transmitted to the Authentication process, which verifies the user's credentials and ensures secure access to the system. Simultaneously, the service request data is sent to the Booking process, where the request is logged, and relevant details are forwarded to the Admin entity for administrative oversight.

4.6.3 Worker DFD Level 1

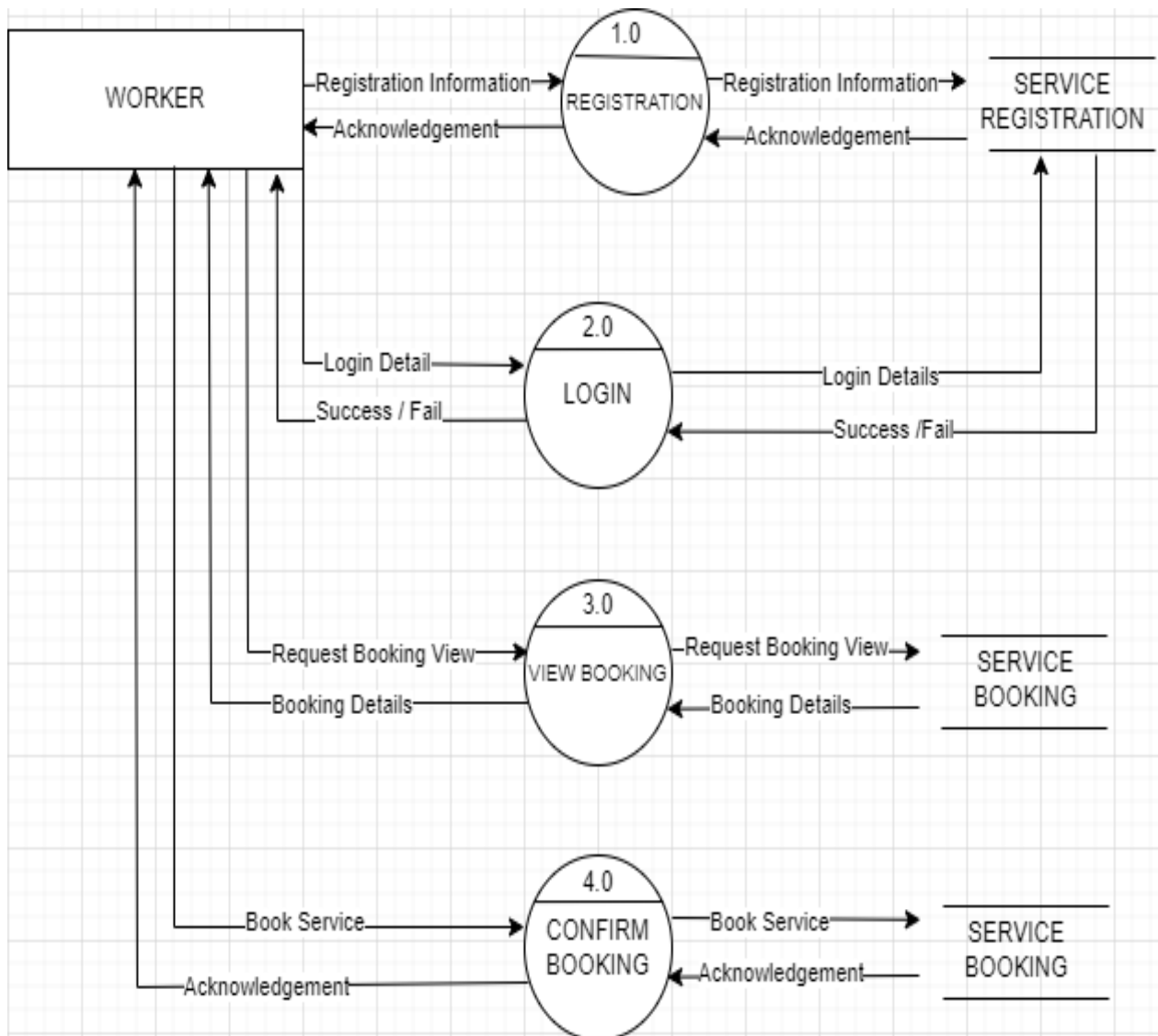


Figure 4.8 Worker DFD Level 1

For the Worker entity, the DFD Level 1 outlines the process from receiving service requests to delivering services. Service requests flow to the Authentication process to ensure that only authenticated and verified workers can access the system. Authenticated workers then receive service requests through the Worker Interface, where they can accept or decline based on their availability and expertise. Accepted requests are processed through the Booking system, notifying the Admin entity of the successful match and updating the User entity on the service status.

4.6.4 Admin DFD Level 1

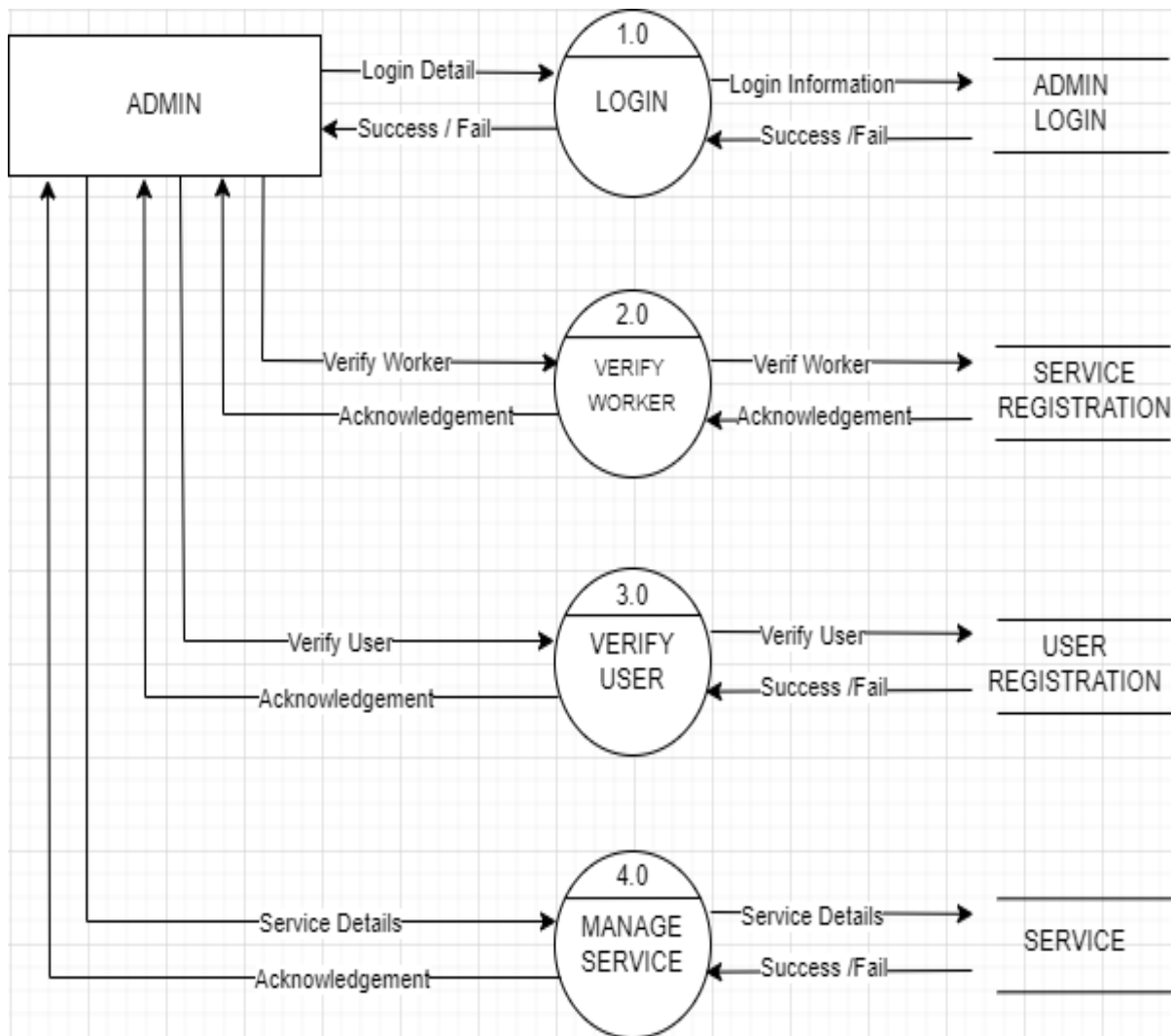


Figure 4.9 Admin DFD Level 1

The DFD Level 1 for the Admin entity delineates the comprehensive role of administrators in managing bookings and ensuring system functionality. Admins receive service request information from the User and Worker entities. They play a pivotal role in coordinating the booking process by confirming service requests, assigning tasks to available workers, and updating users on service status. Admins interact with the Authentication process to ensure secure access, and they contribute to the overall system efficiency by overseeing and optimizing service assignments based on worker availability and user preferences. This detailed breakdown at DFD Level 1 provides a nuanced understanding of the intricate interactions within each entity of the Sahulat App.

Chapter 5

Implementation

5. Implementation

In the implementation phase of the Sahulat App, a dedicated team has translated the conceptual design into a functional and user-friendly application. Leveraging cutting-edge technologies and adhering to the incremental development model, the team has brought the app to life, focusing on user-centricity, security, and efficiency. The implementation phase involves the integration of frontend and backend components, ensuring a seamless experience for both users and workers.

5.1 Algorithm

In the implementation phase of the Sahulat App, we have developed algorithms to facilitate various functionalities of the application. These algorithms govern the core processes, ensuring smooth interactions between users, workers, and administrators. The User Registration and Booking Algorithm ensures a secure and streamlined onboarding process. The Worker Interaction Algorithm optimizes task management for workers, offering a systematic approach to handling service requests.

- i. **User Registration and Booking Algorithm:** This algorithm manages the user registration process and allows users to book services seamlessly. It ensures that user data is securely stored and that booking requests are efficiently processed.
- ii. **Worker Interaction Algorithm:** For worker, this algorithm handles interactions related to service requests, offering a streamlined workflow for accepting or declining requests. It optimizes worker efficiency by providing a systematic approach to managing work and responding promptly to booking requests.
- iii. **Admin Management Algorithm:** The admin module is equipped with an algorithm that allows administrators to oversee and manage all activities within the app. Administrators can seamlessly add, edit, or remove workers, ensuring a dynamic and responsive platform. This algorithm contributes to the efficient functioning of the Sahulat App, allowing administrators to maintain an up-to-date and well-organized system that aligns with the evolving needs of users and workers.

Together, these algorithms form the backbone of the Sahulat App, promising a secure, efficient, and user-centric solution for accessing essential services.

5.2 User Interface

Below figures represents the user interface and screens of Sahulat App. Figure 5.1 shows App Splash Screen of Sahulat App, Figure 5.2 shows User Selection Screen of Sahulat App, Figure 5.3 shows User Homepage of Sahulat App, Figure 5.4 shows Service Details Screen of Sahulat App, Figure 5.5 shows Booking Form Screen of Sahulat App, Figure 5.6 shows Booking Details Screen of Sahulat App, Figure 5.7 shows Worker Home Screen of Sahulat App, Figure 5.8 shows Admin Login of Sahulat App, Figure 5.9 shows Admin Authentication Screen of Sahulat App, Figure 5.10 shows Admin Homepage of Sahulat App.



Figure 5.1 App Splash Screen of Sahulat App

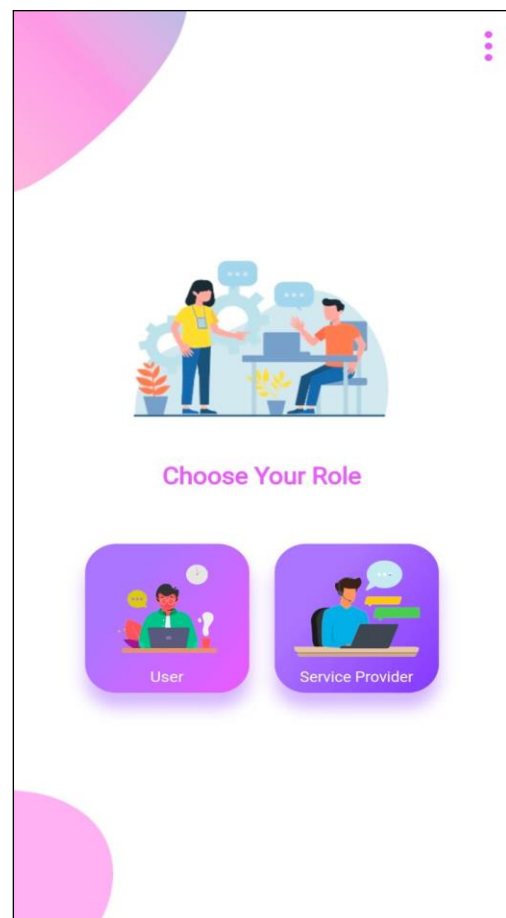


Figure 5.2 User Selection Screen of Sahulat App

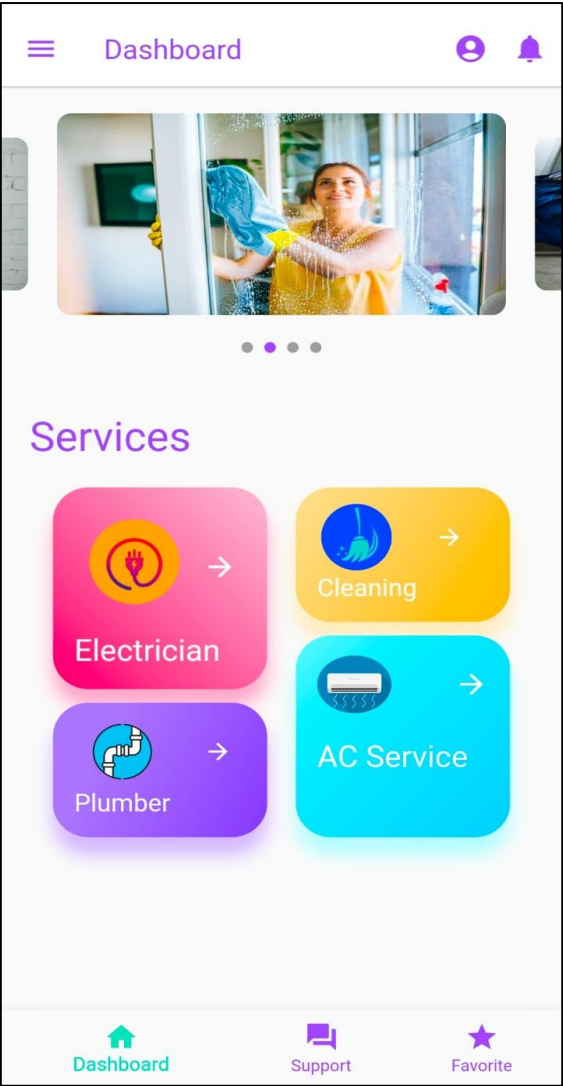


Figure 5.3 User Homepage of Sahulat App

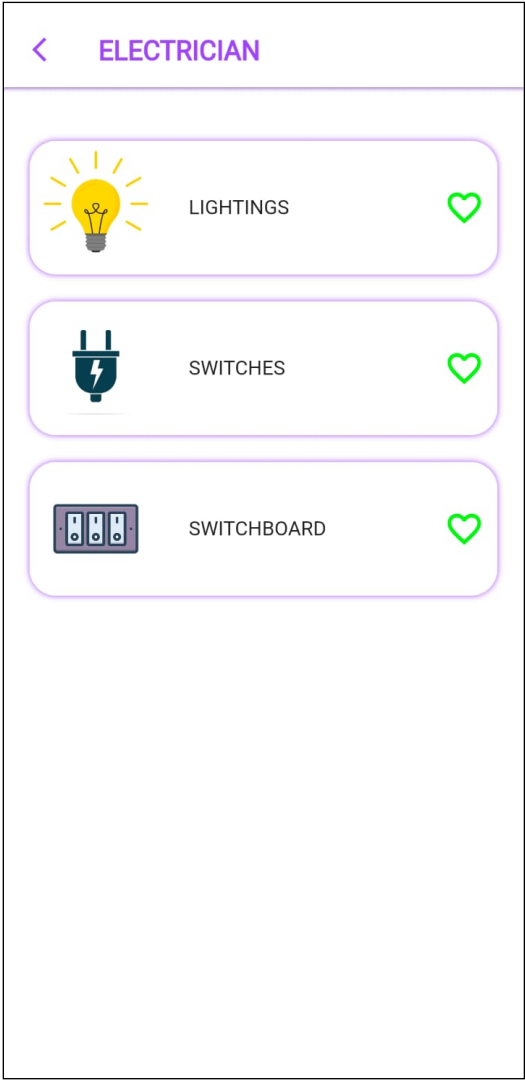


Figure 5.4 Service Details Screen of Sahulat App

< Booking Form

Pick Service Provider

Book Now Not Selected

Pick Date & Time

Date Time

Personal Details

Name

Arslan Nasim

Booking Address

GT ST# 321

Contact Number

00000000000

Book Now

Figure 5.5 Booking Form Screen of Sahulat App

Booking Details

Your details

Name : Arslan Nasim

Contact : 000000000000

Service details

Type : Electrician

Name : LIGHTINGS

Price : Rs10

Provider details

Name : Arslan Nasim

Rating : 3.0

Contact : 03168683835

Booking address

GT ST# 321

Booking slot

2023-10-09 8:56 AM

Cancel Confirm

Book Now

Figure 5.6 Booking Details Screen of Sahulat App

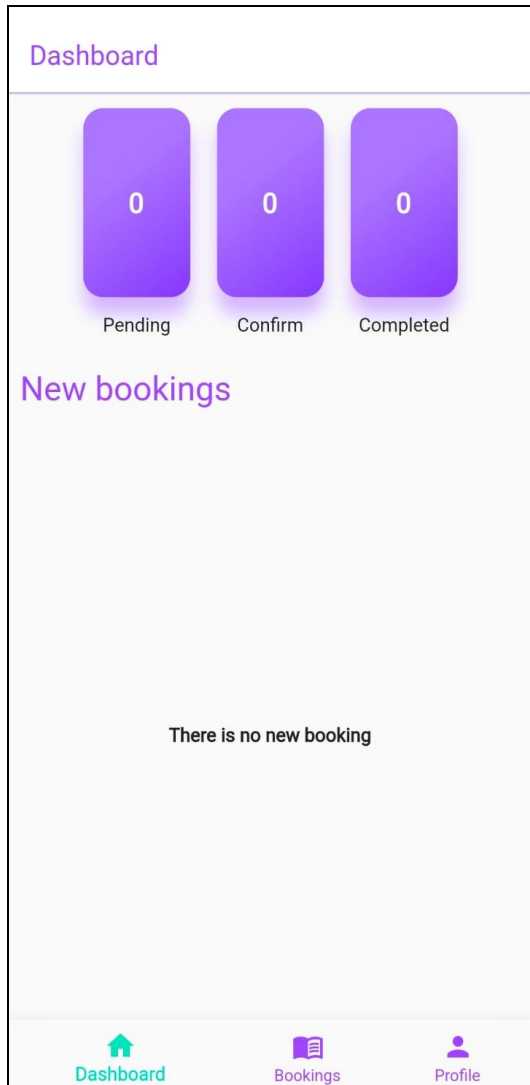


Figure 5.7 Worker Home Screen of Sahulat App

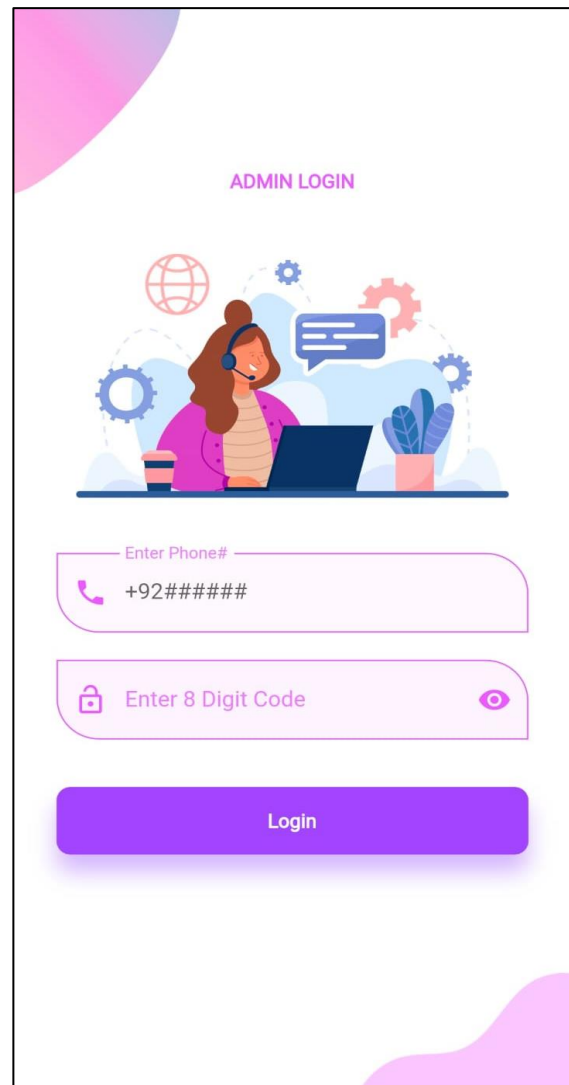


Figure 5.8 Admin Login of Sahulat App

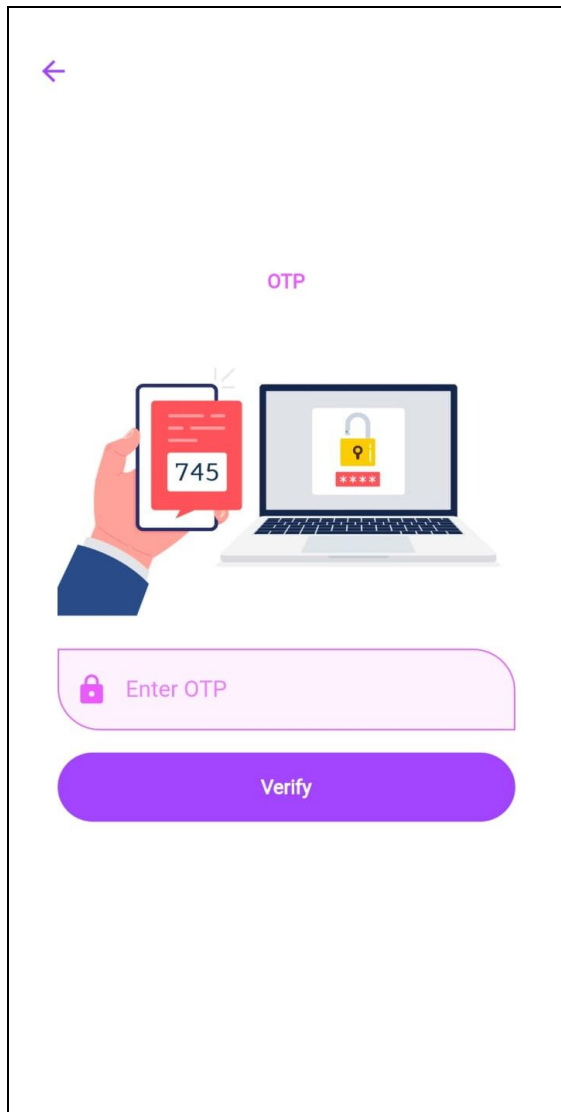


Figure 5.9 Admin Authentication Screen of Sahulat App

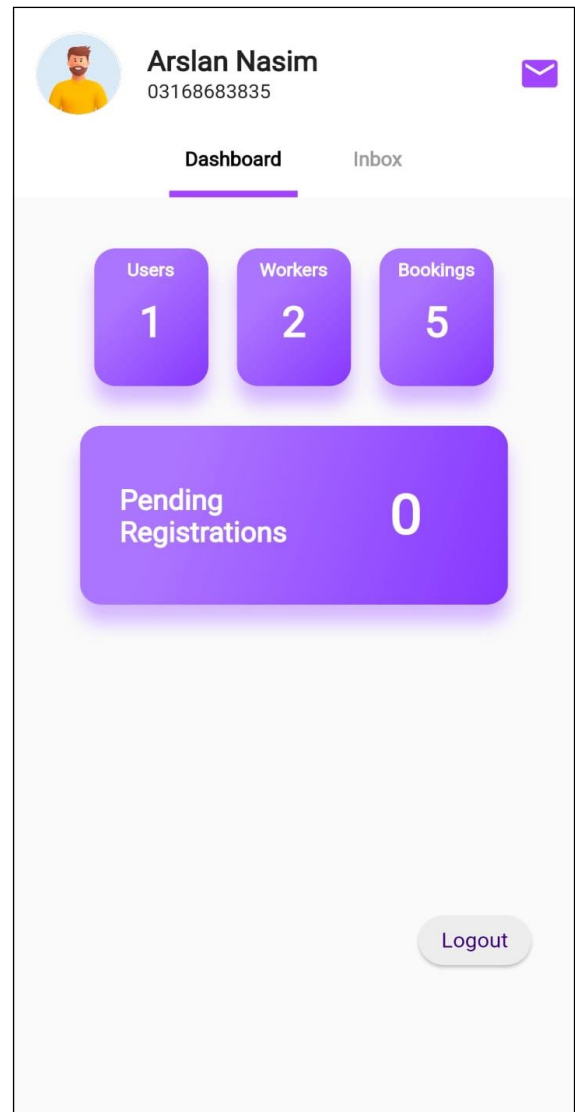


Figure 5.10 Admin Homepage of Sahulat App

Chapter 6

Testing and Evaluation

6. Testing

Testing and evaluation are critical phases in the development of the Sahulat App project to ensure that the platform operates as expected, is secure, and provides an excellent user experience. Here, we outline the key aspects of testing and evaluation:

6.1 Manual Testing

6.1.1 System Testing

When our system has been successfully developed, testing must be performed to ensure that the system is working properly. This is also to check that the system meets the requirements stated earlier. Besides that, system testing will help in finding the errors that may be hidden from the user.

6.1.2 Unit Testing

Unit tests are conducted to assess the individual components of the Sahulat App, such as user login, worker login etc. These tests focus on verifying the correctness and reliability of each component.

Table 6.1 Unit Testing

No.	Test case/ Test Script	Attribute and value	Expected result	Actual result
1.	Verify user login after entering the wrong credentials	Email: emaslanz@gmail.com Password: 123	Login failed	Pass
2.	Verify user login after entering the right credentials	Email: emaslanz@gmail.com Password: 123456	Login successful	Pass
3.	Verify worker after entering the right credentials	Email: muhammadbilal63400@gmail.com Password: 123456	Login successful	Pass

4.	Verify the worker after entering the wrong credentials.	Email: muhammadbilal63400@gmail.com Password: 1234	Login failed	Pass
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6.1.3 Functional Testing

Functional testing examines the system's functionality as a whole. It includes exploring all primary features, such as user book service and confirm service. The system's ability to maintain privacy, security, and responsiveness is assessed.

Table 6.2 Functional Testing

No.	Test case/ Test Script	Attribute and value	Expected result	Actual result
1	User book service	Pick a time and date. Add personal details	booked	Pass
2	Service provides confirmed booking	Booking list Confirm booking	confirmed	pass

6.1.4 Integration Testing

Integration testing involves evaluating the seamless collaboration and functionality of individual components when integrated into a unified system. During this phase, different modules, such as user registration, worker interactions, and admin management, are systematically combined and tested to ensure they interact harmoniously.

Table 6.3 Integration Testing

No.	Test case/ Test Script	Attribute and value	Expected result	Actual result
1	Test and configure the database connection	Firebase real-time connection	Connection Established	pass

Chapter 7

Conclusion and Future Work

7. Conclusion and Future Work

7.1 Conclusion

The Sahulat App represents a significant advancement in addressing the challenges faced in accessing daily life services. The app offers a convenient and user-friendly platform that bridges the gap between service seekers and providers. It not only caters to household customers but extends its services to hospitals, organizations, and offices, providing a comprehensive solution.

The system aims to enhance the overall experience of users by offering secure and accessible services. Additionally, it provides an opportunity for skilled workers to register as workers, expanding the workforce and employment opportunities. Sahulat App use technology to simplify service booking, communication, and feedback, ensuring a seamless experience for all users.

As the app targets the Sahiwal region, it focuses on catering to the local community's specific needs. With its user-centric design and efficient algorithms, the Sahulat App will revolutionize the way daily life services are accessed and delivered. It brings convenience, transparency, and accessibility to a wide range of users, making it a valuable addition to the service industry.

7.2 Future Work

In the future, the Sahulat App project has several avenues for further development and improvement:

- **Enhanced Features:** For administrators, we will consider adding advanced analytics and reporting tools to gain insights into app usage and service trends. This enhancement aims to empower administrators with a deeper understanding of app usage.
- **Subscription System:** For user, we will consider introduce a subscription or membership system for users, providing premium features or benefits.
- **Scalability:** Continuously optimizing the app's infrastructure to handle increased user traffic and content uploads. This commitment to scalability is integral to maintaining a responsive and efficient platform.

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